

Pricing and Hedging American Options Against an Adversarial Exerciser: Methodology

[Author names]

Draft

Abstract

We compute the seller's risk-based price of a Bermudan put under jump and liquidity risk when the holder's exercise time is chosen adversarially among all non-anticipative rules. The price is the value of a two-player zero-sum stochastic game between a hedging policy (the seller) and a stopping policy (the holder), with the seller's conditional value-at-risk (CVaR) of the hedged loss as payoff. Both policies are neural networks trained by gradient descent-ascent through a differentiable market simulator. This note sets out the model, the pricing problem, its basic properties and no-arbitrage guarantees, the numerical method, and the evaluation protocol, including a martingale-duality certificate for the worst case.

1 Market model

Time grid and information. Exercise and rebalancing take place on the grid $0 = t_0 < t_1 < \dots < t_N = T$ with $t_n = n \Delta t$, $\Delta t = T/N$. At each step $n \rightarrow n+1$ the market draws a vector of shocks $\xi_{n+1} = (Z_{n+1}, N_{n+1}, Z'_{n+1}, \varepsilon_{n+1}, \varepsilon'_{n+1})$, independent over time, with $Z, Z', \varepsilon, \varepsilon' \sim \mathcal{N}(0, 1)$ and $N_{n+1} \sim \text{Poisson}(\lambda \Delta t)$. The information available at t_n is $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$. All decisions are required to be $(\mathcal{F}_n)$ -adapted.

Stock price. The stock follows a Merton jump-diffusion, simulated exactly on the grid:

$$\log S_{n+1} = \log S_n + \left(\mu - \lambda \kappa - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z_{n+1} + N_{n+1} m + \delta_J \sqrt{N_{n+1}} Z'_{n+1}, \quad \kappa = e^{m + \delta_J^2/2} - 1. \quad (1)$$

Prices are expressed in discounted units, $\tilde{S}_n = e^{-rt_n} S_n$. Unless stated otherwise $\mu = r$, so the simulation measure is a pricing measure.

Liquidity state. Three non-tradable state variables are computed from returns $r_n = \log S_n - \log S_{n-1}$ and the liquidity noises, using only information up to t_n :

$$x_{n+1} = \min \left(x_n - \kappa_\ell x_n \Delta t + \eta \sqrt{\Delta t} \varepsilon_{n+1} + \beta (-r_{n+1} - h)^+, \log \ell_{\max} \right), \quad \ell_n = e^{x_n}, \quad x_0 = 0, \quad (2)$$

$$\log v_n = \gamma_v \left(\min(|z_n|, z_{\text{cap}}) - \sqrt{2/\pi} \right) + \zeta \varepsilon'_n - \frac{1}{2} \zeta^2, \quad z_n = \frac{r_n}{\sigma \sqrt{\Delta t}}, \quad (3)$$

$$\hat{\sigma}_n^2 = (1 - w) \hat{\sigma}_{n-1}^2 + w \frac{r_n^2}{\Delta t}, \quad \hat{\sigma}_0 = \sigma, \quad (4)$$

with $h = h_{\text{mult}} \sigma \sqrt{\Delta t}$. Illiquidity ℓ mean-reverts to 1 and jumps up after large declines only; relative volume v rises with the size of price moves; $\hat{\sigma}$ is an exponentially weighted volatility estimate. The state affects trading costs only; it does not feed back into prices.

Trading costs. Trading $q \geq 0$ shares per option at t_n costs, in discounted units,

$$c_n(q) = \left[s_0 \ell_n \rho_n q + Y \ell_n \frac{\rho_n \sigma}{\sqrt{252}} \sqrt{\frac{\psi}{v_n}} q^{3/2} \right] \tilde{S}_n, \quad \rho_n = \min\left(\frac{\hat{\sigma}_n}{\sigma}, \rho_{\max}\right), \quad (5)$$

a volatility-proportional half-spread plus square-root market impact. The cost is convex in q . Settings without liquidity risk use proportional costs $c_n(q) = c q \tilde{S}_n$ instead, with $c \in \{0, 0.005\}$.

2 The pricing problem

Contract. A Bermudan put with discounted payoff $\tilde{Z}_n = e^{-rt_n}(K - S_n)^+$, exercisable at $n = 1, \dots, N$ and exercised automatically at N .

Players.

- The *holder* chooses an exercise rule, a stopping time τ with values in $\{1, \dots, N\}$, i.e. $\{\tau = n\} \in \mathcal{F}_n$. Let $\mathcal{T}$ denote the set of such rules.
- The *seller* chooses a hedge $\delta = (\delta_0, \dots, \delta_{N-1})$ with δ_n $\mathcal{F}_n$ -measurable, held over $[t_n, t_{n+1})$. Let $\mathcal{D}$ denote the set of admissible hedges.

Assumption 1. (i) $|\delta_n| \leq \delta_{\max}$ for all n ; (ii) $\mathbb{E}|\tilde{S}_n| < \infty$ and $\mathbb{E}|\tilde{Z}_n| < \infty$; (iii) costs are non-negative and vanish when nothing is traded.

Seller's loss. With $\delta_{-1} = 0$, define

$$\begin{aligned} \tilde{G}_n(\delta) &= \sum_{k < n} \delta_k (\tilde{S}_{k+1} - \tilde{S}_k) && \text{(hedge gains)} \\ C_n(\delta) &= \sum_{k < n} c_k (|\delta_k - \delta_{k-1}|) && \text{(trading costs)} \\ U_n(\delta) &= c_n (|\delta_{n-1}|) \text{ (cash unwind) or } 0 && \text{(cost of closing the hedge at exercise)} \\ L_n(\delta) &= \tilde{Z}_n - \tilde{G}_n(\delta) + C_n(\delta) + U_n(\delta) && \text{(seller's loss if exercised at } t_n). \end{aligned}$$

Risk measure. For $\alpha \in [0, 1)$ and integrable X ,

$$\text{CVaR}_\alpha(X) = \min_{c \in \mathbb{R}} \left\{ c + \frac{\mathbb{E}[(X - c)^+]}{1 - \alpha} \right\} = \frac{1}{1 - \alpha} \int_\alpha^1 \text{VaR}_u(X) du, \quad (6)$$

the average of the worst $(1 - \alpha)$ share of outcomes; $\text{CVaR}_0 = \mathbb{E}$. We use that CVaR_α is monotone, cash-invariant ($\text{CVaR}_\alpha(X + m) = \text{CVaR}_\alpha(X) + m$), convex, and non-decreasing in α .

Definition 1 (Prices). For an assumed exercise rule $\tau_0 \in \mathcal{T}$ (e.g. the Longstaff–Schwartz rule) and for worst-case exercise,

$$p(\tau_0) = \inf_{\delta \in \mathcal{D}} \text{CVaR}_\alpha(L_{\tau_0}(\delta)), \quad W(\delta) = \sup_{\tau \in \mathcal{T}} \text{CVaR}_\alpha(L_\tau(\delta)), \quad p^* = \inf_{\delta \in \mathcal{D}} W(\delta).$$

The *underpricing* of the assumed rule is $\Delta(\tau_0) = p^* - p(\tau_0)$. The classical Bermudan price is $V_0 = \sup_{\tau \in \mathcal{T}} \mathbb{E}[\tilde{Z}_\tau]$.

By cash-invariance, $\text{CVaR}_\alpha(L_\tau - p^*) = \text{CVaR}_\alpha(L_\tau) - p^*$, so p^* is the smallest premium at which the seller's average result over the worst $(1 - \alpha)$ of scenarios breaks even under any non-anticipative exercise rule.

3 Properties

Proposition 1 (Robustness). *For every $\delta \in \mathcal{D}$ and $\tau \in \mathcal{T}$, $\text{CVaR}_\alpha(L_\tau(\delta)) \leq W(\delta)$.*

Proof. $W(\delta)$ is the supremum over $\mathcal{T}$ and $\tau \in \mathcal{T}$. □

Proposition 2 (Non-negative underpricing). *For every $\tau_0 \in \mathcal{T}$, $p(\tau_0) \leq p^*$, i.e. $\Delta(\tau_0) \geq 0$.*

Proof. For each δ , $\text{CVaR}_\alpha(L_{\tau_0}(\delta)) \leq W(\delta)$ by Proposition 1; take the infimum over δ . The Longstaff–Schwartz rule decides at t_n from S_n only, hence lies in $\mathcal{T}$. □

Lemma 1 (Martingale property). *If $\mu = r$, then $\mathbb{E}[\tilde{S}_{n+1} \mid \mathcal{F}_n] = \tilde{S}_n$.*

Proof. By independence of the shocks, $\mathbb{E}[S_{n+1}/S_n \mid \mathcal{F}_n] = e^{(r - \lambda\kappa - \sigma^2/2)\Delta t} \mathbb{E}[e^{\sigma\sqrt{\Delta t}Z}] \mathbb{E}[e^J]$, where J is the jump term in (1). Here $\mathbb{E}[e^{\sigma\sqrt{\Delta t}Z}] = e^{\sigma^2\Delta t/2}$, and J is a $\text{Poisson}(\lambda\Delta t)$ sum of independent $\mathcal{N}(m, \delta_J^2)$ jumps, each with $\mathbb{E}[e^Y] = 1 + \kappa$, so $\mathbb{E}[e^J] = \exp(\lambda\Delta t \kappa)$. Hence $\mathbb{E}[S_{n+1}/S_n \mid \mathcal{F}_n] = e^{r\Delta t}$. □

Proposition 3 (Risk-neutral reduction). *Let $\mu = r$ and $\alpha = 0$. Then $p^* = V_0$, attained by the hedge $\delta \equiv 0$, also in the presence of non-negative trading costs. For every $\tau_0 \in \mathcal{T}$, $\Delta(\tau_0) = V_0 - \mathbb{E}[\tilde{Z}_{\tau_0}] \geq 0$, with equality if τ_0 is a risk-neutral optimal rule.*

Proof. By Lemma 1 and Assumption 1(i), $\tilde{G}$ is a martingale with $\tilde{G}_0 = 0$. For bounded $\tau \leq N$, $\tilde{G}_\tau = \sum_k \mathbf{1}\{\tau > k\} \delta_k(\tilde{S}_{k+1} - \tilde{S}_k)$ with $\mathbf{1}\{\tau > k\}$ $\mathcal{F}_k$ -measurable, so $\mathbb{E}[\tilde{G}_\tau] = 0$ (optional stopping). Hence $\mathbb{E}[L_\tau(\delta)] = \mathbb{E}[\tilde{Z}_\tau] + \mathbb{E}[C_\tau + U_\tau] \geq \mathbb{E}[\tilde{Z}_\tau]$, with equality at $\delta \equiv 0$. Taking $\sup_\tau$ then $\inf_\delta$ gives $p^* = V_0$. Similarly $p(\tau_0) = \mathbb{E}[\tilde{Z}_{\tau_0}]$. □

Proposition 4 (Ordering). *(a) p^* is non-decreasing in α . (b) If $\mu = r$, then $p^* \geq V_0$ for every α . (c) If $x \in \mathbb{R}$ and $\delta \in \mathcal{D}$ satisfy $x + \tilde{G}_n(\delta) - C_n(\delta) - U_n(\delta) \geq \tilde{Z}_n$ for all n almost surely (a super-hedge), then $p^* \leq x$.*

Proof. (a) By (6), CVaR_α averages the non-decreasing map $u \mapsto \text{VaR}_u$ over $[\alpha, 1]$; suprema and infima preserve the ordering. (b) Follows from (a) and Proposition 3. (c) The super-hedge gives $L_n(\delta) \leq x$ for all n , hence $L_\tau(\delta) \leq x$ and, by monotonicity, $W(\delta) \leq x$. □

Proposition 5 (Order of play). *$\sup_\tau \inf_\delta \text{CVaR}_\alpha(L_\tau(\delta)) \leq p^*$. If (δ^*, τ^*) is a saddle point, then $p^* = \text{CVaR}_\alpha(L_{\tau^*}(\delta^*))$ and the two orders of play agree.*

Proof. For any τ', δ' : $\inf_\delta \text{CVaR}_\alpha(L_{\tau'}(\delta)) \leq \text{CVaR}_\alpha(L_{\tau'}(\delta')) \leq W(\delta')$; take $\sup_{\tau'}$ and $\inf_{\delta'}$. At a saddle point both inequalities hold with equality. □

Remark 1. Existence of a saddle point is not guaranteed: the objective is not convex–concave in the network parameters and pure stopping rules do not form a convex set. Relaxed (randomised) stopping, used in training (Section 5), is the standard device that convexifies stopping games. We therefore approximate the seller-first value p^* and assess the approximation empirically (Section 6) and, on the holder's side, rigorously via duality (Section 7).

4 No-arbitrage

Theorem 1 (Arbitrage-free market). *Let $\mu = r$. No admissible strategy δ satisfies $\tilde{G}_N(\delta) - C_N(\delta) \geq 0$ almost surely with $\mathbb{P}(\tilde{G}_N(\delta) - C_N(\delta) > 0) > 0$.*

Proof. By Lemma 1, $\mathbb{E}[\tilde{G}_N] = 0$. If $\tilde{G}_N - C_N \geq 0$ then $\tilde{G}_N \geq C_N \geq 0$; a non-negative variable with zero mean vanishes almost surely, so $\tilde{G}_N = 0$ and then $C_N = 0$. $\square$

Theorem 2 (Arbitrage-free price). *Let $\mu = r$ and let π_{sub} and π_{sup} denote the sub- and super-hedging prices of the option. Then $\pi_{\text{sub}} \leq V_0 \leq p^* \leq \pi_{\text{sup}}$, so trading the option at p^* creates no arbitrage.*

Proof. If a buyer can lock in x , i.e. $-x + \tilde{Z}_\tau + \tilde{G}_\tau(\delta) - C_\tau(\delta) \geq 0$ a.s. for some τ, δ , taking expectations and using $\mathbb{E}[\tilde{G}_\tau] = 0$ and $C \geq 0$ gives $x \leq \mathbb{E}[\tilde{Z}_\tau] \leq V_0$; hence $\pi_{\text{sub}} \leq V_0$. The remaining inequalities are Proposition 4(b),(c). $\square$

Theorem 3 (Consistency across strikes). *Write $p^*(K)$ for the price at strike K . For $K' < K$, $0 \leq p^*(K) - p^*(K') \leq K - K'$, and $K \mapsto p^*(K)$ is convex.*

Proof. Pathwise $0 \leq (K - S)^+ - (K' - S)^+ \leq K - K'$, so $L(K') \leq L(K) \leq L(K') + (K - K')$ for every δ, τ ; monotonicity and cash-invariance carry this through CVaR, $\sup_\tau$ and $\inf_\delta$. For convexity, $L_n(\delta, K)$ is jointly convex in (δ, K) (payoff convex in K , gains linear in δ , costs convex in δ); CVaR is convex and monotone, a supremum of convex functions is convex, and partial minimisation of a jointly convex function over δ preserves convexity in K . $\square$

Remark 2. These results concern admissible strategies. The implementation enforces adaptedness through the network inputs (checked by permutation tests on future shocks) and should bound the hedge output, e.g. $\delta_n = \delta_{\max} \tanh(\cdot)$. Under a real-world measure ($\mu \neq r$), hedge gains carry a risk premium, and the seller's price must be defined as an indifference price, $p^* - \inf_\delta \text{CVaR}_\alpha(-\tilde{G}_N(\delta) + C_N(\delta))$, to exclude speculative gains.

5 Numerical method

Policies. Both players are feed-forward networks (two hidden layers of width 64, SiLU activations) shared across dates, with time as an input. Inputs are rescaled by fixed constants ($\sigma\sqrt{T}$ for log-moneyness, $K\sigma\sqrt{T}$ for money amounts):

$$\begin{aligned}\delta_n &= H_\theta(t_n/T, \log(S_n/K), \delta_{n-1}, \log \ell_n, \log v_n, \log(\hat{\sigma}_n/\sigma)), \\ g_n &= S_\phi(t_n/T, \log(S_n/K), \delta_{n-1}, L_n, \mathbf{1}\{S_n < K\}, \log \ell_n, \log v_n, \log(\hat{\sigma}_n/\sigma)).\end{aligned}$$

Both are $\mathcal{F}_n$ -measurable by construction. The liquidity inputs are omitted in settings without liquidity risk.

Relaxed stopping. Training uses stopping probabilities $f_n = \text{sigmoid}(g_n/T_{\text{temp}})$ with $f_N = 1$, the stopping distribution $p_n = f_n \prod_{k < n} (1 - f_k)$, and the per-path relaxed loss $X = \sum_n p_n L_n$. The temperature is annealed geometrically from 1 to 0.1. All reported values use the hard rule $\tau = \min\{n : g_n > 0\} \wedge N$, which is a stopping time.

Objective and gradients. On a batch of B simulated paths, CVaR_α is estimated by the mean of the largest $\lceil (1 - \alpha)B \rceil$ values of X and differentiated through the selection. Gradients flow pathwise through the simulator with respect to the network parameters; the shocks are fixed given the draw and do not depend on the policies (deep hedging; deep empirical risk minimisation). The hedger’s gradient also flows through the stopper’s inputs (δ_{n-1}, L_n) , so the hedger responds to the holder’s reaction.

Gradient descent–ascent. Each outer iteration performs $k = 5$ ascent steps on ϕ (holder, maximising the relaxed CVaR) with fresh paths and the hedger frozen, then one descent step on θ (seller). Adam with learning rate 10^{-3} and cosine decay is used for both. Stabilisers:

- *Warm start:* the stopper is first fitted (300 binary cross-entropy steps) to the rule “exercise iff $S_n < K$ and $K - S_n$ exceeds the Merton European value”, since a randomly initialised stopper collapses to never exercising early.
- *Adversary pool:* every 200 iterations a copy of the stopper is stored; the hedger minimises the maximum of the CVaR against the current stopper and against a randomly drawn stored copy.
- *Restarts* (used at $\alpha = 0.99$): every 150 iterations a freshly warm-started stopper is trained briefly and replaces the current one if it attacks the hedger more strongly.

Baselines. (i) The *Longstaff–Schwartz* rule: continuation values regressed on a cubic polynomial in $S/K - 1$ over in-the-money paths, fitted on 2^{18} paths and priced on an independent set of 2^{18} paths. (ii) A *CRR* binomial tree at $\lambda = 0$ (American and Bermudan-on-grid). (iii) Fixed-rule hedgers trained against the Longstaff–Schwartz rule and against holding to maturity, whose CVaR under the assumed rule estimates $p(\tau_0)$.

6 Evaluation protocol

- All reported numbers are computed on a fixed held-out test set of 2^{18} paths, never on training batches.
- Standard errors of CVaR_α use the influence function $\psi(x) = \text{VaR}_\alpha + (x - \text{VaR}_\alpha)^+ / (1 - \alpha)$; differences between hedges or rules are computed paired on the same paths.
- *Exploitability:* each frozen hedger is attacked by a freshly initialised stopper (2000 steps) and by simple rules (never early, Longstaff–Schwartz, running-loss threshold, illiquidity threshold, post-decline). The gap to the co-trained adversary measures how far the holder is from a best response.
- *Validation:* at $\alpha = 0$ the method must recover the Longstaff–Schwartz and CRR prices (Proposition 3); decisions must be unchanged when future shocks are permuted.
- *Sanity checks:* an option with zero payoff prices to 0; the sample mean of $\tilde{S}_{n+1} - \tilde{S}_n$ is zero within standard error, including conditionally on all policy inputs; prices across strikes satisfy Theorem 3.

7 Martingale-duality certificate

For a fixed hedge δ , the trained adversary yields a lower estimate of $W(\delta)$. An upper bound follows from martingale duality (Rogers 2002; Haugh and Kogan 2004).

Theorem 4 (Dual upper bound). *Fix $\delta \in \mathcal{D}$. For each $c \in \mathbb{R}$ let M^c be an $(\mathcal{F}_n)$ -martingale with $M_0^c = 0$ and integrable increments. Then*

$$W(\delta) \leq D(\delta) = \inf_{c \in \mathbb{R}} \left\{ c + \frac{1}{1 - \alpha} \mathbb{E} \left[\max_{1 \leq n \leq N} ((L_n(\delta) - c)^+ - M_n^c) \right] \right\}.$$

Proof. By (6), $W(\delta) = \sup_{\tau} \min_c F(\tau, c)$ with $F(\tau, c) = c + \mathbb{E}[(L_{\tau} - c)^+]/(1 - \alpha)$, and $\sup \min \leq \min \sup$. For fixed c , optional stopping gives $\mathbb{E}[M_{\tau}^c] = 0$, so $\mathbb{E}[(L_{\tau} - c)^+] = \mathbb{E}[(L_{\tau} - c)^+ - M_{\tau}^c] \leq \mathbb{E}[\max_n ((L_n - c)^+ - M_n^c)]$ for every $\tau \in \mathcal{T}$. $\square$

Hence, for each hedge, $\text{CVaR}_{\alpha}(L_{\hat{\tau}}(\delta)) \leq W(\delta) \leq D(\delta)$, where $\hat{\tau}$ is the trained adversary. Martingales are built from shocks with known zero conditional mean, e.g. $\Delta M_{n+1}^c = \sum_k \varphi_k(\mathcal{F}_n) \xi_{n+1}^{(k)}$ with $\xi \in \{Z, Z^2 - 1, N - \lambda \Delta t, \varepsilon, \varepsilon'\}$, where φ is a network trained to minimise D ; every choice gives a valid bound. The bound certifies the worst case for a given hedge and yields a certified upper bound $p^* \leq D(\hat{\delta})$; it does not bound p^* from below, since p^* is an infimum over all hedges. Two sources of slack remain: the sup-min exchange (CVaR is not time-consistent) and the inability of stock-driven martingales to offset jumps. The threshold c should be selected on paths independent of those used to evaluate the bound.

8 Parameters

Group	Parameter	Value
Contract	S_0, K, T, N	100, 100, 0.5, 50
Market	$r, \sigma, \lambda, m, \delta_J$	0.03, 0.20, 1, -0.10, 0.15
Liquidity	$\kappa_{\ell}, \eta, \beta, h_{\text{mult}}, \ell_{\text{max}}$	20, 1, 15, 2, 10
	$\gamma_v, \zeta, z_{\text{cap}}, w$	0.3, 0.2, 10, 0.3
Costs	$s_0, Y, \psi, \rho_{\text{max}}$	0.0005, 1, 0.05, 4
Risk	α	0, 0.5, 0.9, 0.95, 0.99 (headline 0.9)
Training	batch, outer iterations	2^{12} , 1500 (headline); 600 (grid)
	ascent steps per descent step, learning rate	5, 10^{-3}
Evaluation	test paths, exploit steps	2^{18} , 2000

References

- Andersen, L. and Broadie, M. (2004). Primal-dual simulation algorithm for pricing multidimensional American options. *Management Science* 50(9), 1222–1234.
- Becker, S., Cheridito, P. and Jentzen, A. (2020). Pricing and hedging American-style options with deep learning. *Journal of Risk and Financial Management* 13(7), 158.
- Buehler, H., Gonon, L., Teichmann, J. and Wood, B. (2019). Deep hedging. *Quantitative Finance* 19(8), 1271–1291.
- Haugh, M. and Kogan, L. (2004). Pricing American options: a duality approach. *Operations Research* 52(2), 258–270.

Longstaff, F. and Schwartz, E. (2001). Valuing American options by simulation: a simple least-squares approach. *Review of Financial Studies* 14(1), 113–147.

Reppen, A. M., Soner, H. M. and Tissot-Daguette, V. (2023). Neural optimal stopping boundary. arXiv:2205.04595.

Roch, A. F. (2022). Hedging of American options in illiquid markets with price impacts. *International Journal of Theoretical and Applied Finance*.

Rockafellar, R. T. and Uryasev, S. (2000). Optimization of conditional value-at-risk. *Journal of Risk* 2(3), 21–41.

Rogers, L. C. G. (2002). Monte Carlo valuation of American options. *Mathematical Finance* 12(3), 271–286.